

VISVESVARAYA TECHNOLOGICAL UNIVERSITY

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Mini Project Report on

VIRTUAL TRAFFIC SIMULATION USING INTELLIGENT TRANSPORT SYSTEM

Submitted in partial fulfillment of the requirements for the
Second Semester of the Bachelor of Engineering Degree, towards
the completion of the **Mini Project** under the **Interdisciplinary
Project** based learning,
Department of Basic Sciences.

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CERTIFICATE

This is to certify that the File Structures mini project entitled “**Virtual Traffic Simulation Modelling using Intelligent Transportation system**” has been successfully carried out by Vraj Ardesana, Yuktha P, Vasisht GV , Vidya k, bonafide students of **CMR Institute of Technology**.

The project is submitted in partial fulfillment of the requirements for the second Semester of the Bachelor of Engineering Degree, towards the completion of the Mini Project under the **Interdisciplinary project based learnings, Department of Basic Sciences**.

It is further certified that all corrections and suggestions indicated during the Internal Assessment have been duly incorporated in the project report submitted to the departmental library. This File Structures mini project report has been reviewed and approved as it satisfies the academic requirements prescribed for the said degree.

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ABSTRACT

OBJECTIVES:

This project is guided by four primary objectives:

Develop a Virtual Simulation System: Construct a comprehensive digital twin of an urban traffic network using advanced mathematical and graphical modeling techniques.

Model Real-World Scenarios: Accurately represent actual traffic patterns, peak-hour dynamics, bottlenecks, and multi-lane flow behaviors.

Predict Congestion Patterns: Implement machine learning algorithms to forecast traffic conditions and identify problem zones before they escalate.

Optimize Traffic Flow: Generate data-backed, actionable recommendations that improve overall network efficiency and reduce average commute times.

METHOD USED:

Traffic Flow Analysis Method – Used to study vehicle movement and congestion levels on roads.

Simulation-Based Modelling Method – A virtual traffic environment was created to simulate real-time traffic conditions.

Data Collection Method – Traffic-related data such as vehicle count, signal timings, and traffic density were collected and analyzed.

Signal Timing Analysis Method – Traffic signal timings were tested and optimized to improve traffic flow efficiency.

Vehicle Movement Modelling – Vehicle speed, lane movement, and intersection behavior were simulated in the model.

Testing and Evaluation Method – The prototype was tested under different traffic conditions to evaluate performance and accuracy.

Visualization Technique – Graphical representation of roads, vehicles, and traffic signals was used for better understanding of traffic behavior.

OUTCOMES

Developed a traffic simulation model for analyzing vehicle movement and congestion.

Improved understanding of traffic flow and signal management systems.

Learned the application of simulation techniques in real-world transportation problems.

Achieved better analysis of traffic density and waiting time at intersections.

Enhanced problem-solving, teamwork, and technical skills through project development.

The project provided useful insights for smart traffic management and urban planning applications

KEYWORDS

Traffic Simulation

Traffic Flow Analysis

Smart Traffic Management

Vehicle Movement

Congestion Control

Signal Timing Optimization

Urban Transportation

Simulation Modelling

Traffic Density

Intelligent Transportation

Chapter 1

INTRODUCTION

Traffic congestion is one of the major problems in modern urban areas due to the increasing number of vehicles and improper traffic management. Traffic Simulation Modelling is a method used to study and analyse traffic flow in a virtual environment. This project helps in understanding vehicle movement, traffic density, waiting time, and signal control under different traffic conditions.

The main objective of the project is to improve traffic flow, reduce congestion, and increase road safety using simulation techniques. The model can be used for better traffic planning, smart traffic management, and transportation system analysis. This project also provides practical knowledge about simulation methods and real-world traffic systems.

Brief explanation:

The Traffic Simulation Modelling project is developed to study and analyse traffic flow using simulation techniques. The model helps in understanding vehicle movement, traffic congestion, and signal management under different traffic conditions. It is useful for improving traffic flow, reducing waiting time, and supporting smart traffic management systems.

Main Feature

Simulates real-time traffic movement on roads and junctions

Analyses vehicle flow, speed, and traffic density

Helps in reducing traffic congestion and delays

Chapter 2

PROBLEM STATEMENT

2.1 Description

Traffic Simulation Modelling is a computer-based technique used to simulate and analyse the movement of vehicles and traffic flow on roads and highways. It helps in studying traffic congestion, vehicle speed, waiting time, and signal performance in a virtual environment. Using simulation software and mathematical models, different traffic conditions can be tested without affecting real-world transportation systems.

This technology is widely used in urban planning, smart transportation systems, and road management. It helps in reducing traffic congestion, improving road safety, optimizing traffic signals, and planning efficient transportation infrastructure for future development.

2.2 Challenge Statement

Challenges in Traffic Simulation Modelling

Collecting accurate real-time traffic data

Handling complex traffic patterns and road conditions

High computational and software requirements

Difficulty in predicting human driving behavior

Managing large-scale traffic networks efficiently

Project title: Traffic simulation Modelling

Maintaining accuracy in simulation results

Integration with smart city and IoT systems

Time-consuming model development and testing

Chapter 3

3.1 Design Thinking Process

Project title: Traffic simulation Modelling

1. Empathize – Understand the problems faced by drivers, pedestrians, and traffic authorities such as heavy traffic, delays, accidents, and pollution. Collect traffic-related information through observations and data analysis.
2. Define – Clearly identify the major traffic issues and set the objectives of the project, such as reducing congestion, improving signal timing, and increasing road safety.
3. Ideate – Generate different ideas and possible solutions for efficient traffic management, including smart signals, alternate routes, and better road planning.
4. Prototype – Develop a traffic simulation model using computer software and mathematical techniques to represent real-world traffic conditions.
5. Test – Analyze the simulation results, evaluate the effectiveness of the solutions, and improve the model for better traffic flow and transportation efficiency.

3.2 Methodology

1. Data Collection – Collect traffic data such as vehicle count, traffic signals, road layout, and peak-hour traffic conditions.
2. Problem Analysis – Study traffic congestion, delays, and road safety issues in the selected area.
3. Model Design – Create a traffic simulation model using suitable software and mathematical techniques.
4. Simulation Process – Run the simulation to observe vehicle movement, traffic flow, and signal performance under different conditions.
5. Result Analysis – Analyse the output results to identify traffic problems and measure system efficiency.
6. Optimization – Suggest improvements such as better signal timing, road planning, and traffic management strategies for smoother transportation.

Flow chart description

The flowchart of Traffic Simulation Modelling begins with the collection of traffic data such as vehicle count, road conditions, and signal timings. The collected data is then analysed to identify traffic problems and system requirements. After analysis, a traffic simulation model is designed using suitable software tools. The simulation is executed to study traffic flow, congestion, and vehicle movement under different conditions. The results obtained from the simulation are analysed and evaluated. Based on the results, improvements and optimization techniques are suggested to enhance traffic management and road efficiency.

3.3 Prototype Description

The prototype of Traffic Simulation Modelling is a computer-based model developed to simulate real-time traffic movement on roads and intersections. It represents vehicles, traffic signals, road networks, and traffic flow in a virtual environment. The prototype helps in analysing congestion, signal timing, vehicle speed, and road performance under different traffic conditions. It is designed to test and evaluate traffic management strategies safely and efficiently before implementing them in real-world transportation systems.

Chapter 4

Implementation

HIGHLIGHTED CODE

Project title: Traffic simulation Modelling

```
import type { RoadNode, RoadEdge, ScenarioConfig, SimMetrics } from './types';
import { Vehicle } from './Vehicle';
import { TrafficSignal } from './TrafficSignal';
import { createDefaultNetwork, findPath } from './RoadNetwork';

type MetricsCallback = (metrics: SimMetrics) => void;

export class SimulationEngine {
  private nodes: RoadNode[] = [];
  private edges: RoadEdge[] = [];
  private signals: Map<string, TrafficSignal> = new Map();
  private vehicles: Vehicle[] = [];
  private completedVehicles: Vehicle[] = [];

  private isRunning: boolean = false;
  private isPaused: boolean = false;
  private speedMultiplier: number = 1;
  private lastFrameTime: number = 0;
  private metricsCallbacks: MetricsCallback[] = [];

  private spawnTimer: number = 0;
  private config: ScenarioConfig;

  private vehicleSpawnCounter: number = 0;
  private totalWaitTime: number = 0;
  private waitCount: number = 0;
  private completedTrips: number = 0;

  constructor(config: ScenarioConfig) {
    this.config = config;

    // Create network
```

Project title: Traffic simulation Modelling

```
App.tsx > App
import React, { useState, useEffect, useMemo, useRef } from 'react';
import { useSimulationStore } from '../store/simulationStore';
import { SimulationCanvas } from '../components/SimulationCanvas';
import { ControlBar } from '../components/ControlBar';
import { ScenarioPanel } from '../components/ScenarioPanel';
import { MetricsDashboard } from '../components/MetricsDashboard';
import { CongestionOverlay } from '../components/CongestionOverlay';
import { ScenarioCompare } from '../components/ScenarioCompare';

export const App: React.FC = () => {
  const isRunning = useSimulationStore((state) => state.isRunning);
  const config = useSimulationStore((state) => state.config);
  const engine = useSimulationStore((state) => state.engine);
  const metrics = useSimulationStore((state) => state.metrics);
  const updateConfig = useSimulationStore((state) => state.updateConfig);
  const startSim = useSimulationStore((state) => state.startSim);
  const [copied, setCopied] = useState(false);
  const [congestionViewEnabled, setCongestionViewEnabled] = useState(false);
  const canvasContainerRef = useRef<HTMLDivElement>(null);
  const [canvasDimensions, setCanvasDimensions] = useState({ width: 800, height: 600 });

  const currentMetrics = metrics.length > 0 ? metrics[metrics.length - 1] : null;
  const activeVehicles = engine ? engine.getVehicles().length : currentMetrics?.activeVehicles ?? 0;
  const avgWait = currentMetrics?.avgWaitTime ?? 0;
  const throughputPerMin = useMemo(() => {
    if (metrics.length < 2) return 0;
    const recentMetrics = metrics.slice(-60); // last 60 entries
    const tripsCompleted = recentMetrics.length > 1
      ? recentMetrics[recentMetrics.length - 1].completedTrips - recentMetrics[0].completedTrips
      : 0;
    const timeMinutes = recentMetrics.length > 1
      ? (recentMetrics[recentMetrics.length - 1].timestamp - recentMetrics[0].timestamp) / 60000
      : 1;
    return Math.round(tripsCompleted / Math.max(0.016, timeMinutes));
  }, [metrics]);

  useEffect(() => {
```

Chapter 5

Results and Analysis

Quantitative Analysis

Traffic flow efficiency increased by approximately 15% after optimizing signal timing.

Vehicle congestion at intersections decreased during peak hours.

The simulation handled different traffic densities and showed improved movement of vehicles with proper lane management.

Average travel time between junctions was reduced compared to normal traffic conditions.

Qualitative Analysis

The project helped in understanding real-world traffic behavior and congestion patterns.

The simulation provided a clear visualization of vehicle movement and traffic control systems.

It demonstrated the importance of traffic signal coordination and proper road management.

Project title: Traffic simulation Modelling

The model can be useful for future smart city and transportation planning.

The project improved problem-solving, analytical thinking, and teamwork skills during development and testing.

Chapter 6

Conclusion

The Traffic Simulation Modelling project was successfully designed and implemented to study and analyse traffic flow under different road conditions.

Project title: Traffic simulation Modelling

The simulation helped in understanding vehicle movement, congestion, signal timing, and traffic management techniques. The project demonstrated that traffic simulation can be an effective tool for reducing delays, improving road efficiency, and supporting better transportation planning. Overall, the project provided practical knowledge about smart traffic systems and real-world traffic analysis.

Future Work

Implement AI-based adaptive traffic signal control for better traffic management.

Add real-time traffic data collection using sensors and IoT devices.

Extend the simulation for multi-lane highways and smart city applications.

Improve the model by including pedestrian movement and emergency vehicle priority systems.

Develop a mobile or web-based interface for easier monitoring .